

GNN-BERT for Music Context Understanding

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Abstract. Music understanding combines acoustical structure with semantic descriptions. This project implements the complete CSE425 GNN-BERT music-context pipeline on MusicCaps: a DistilBERT caption tagger, a GraphSAGE segment-graph encoder, CNN and unimodal baselines, GNN-BERT fusion ablations, and dual-encoder contrastive retrieval. The full locally available corpus was processed into 5,156 audio feature caches and segment graphs, with deterministic 3,608/773/774 train/validation/test splits. Under a five-epoch, time-constrained training budget, early concatenation achieved the strongest held-out mean AUC-PR (0.134), followed by BERT-only (0.138) and the audio graph model (0.125); fixed-threshold F1 remained zero because sparse proxy tags produced probabilities below 0.5. The report therefore emphasizes threshold-free AUC-PR and retrieval Recall@K, and documents limitations and reproducibility artifacts.

1. Introduction

Musical context is not captured by one signal alone. Audio conveys pitch, timbre, rhythm, and repeated sections; natural-language captions identify instruments, performance qualities, and mood. This work tests whether graph structure derived from audio segments complements contextual text embeddings. The implementation follows the assignment's four-task roadmap: textual multi-label tagging, graph-based context prediction, GNN-BERT fusion, and caption-to-audio contrastive retrieval.

2. Data and Preprocessing

MusicCaps provides expert captions for short music clips. A local corpus of 5,156 downloaded waveforms was resampled to 22.05 kHz. We extracted 64-bin log-mel features (1,024-point FFT, 512-sample hop), applied per-track z-normalization, and split clips into two-second windows. The resulting fixed-window features form graph nodes. Temporal neighbors are connected, and additional undirected edges join windows whose mean log-mel cosine similarity exceeds 0.70. Each graph is capped at 12 nodes.

MusicCaps captions do not provide a conventional music-tag matrix. We therefore create a transparent proxy target from the 20 most frequent annotated caption aspects (for example, instrumentation and vocal descriptors). This should be interpreted as a caption-aspect prediction task, not as a standard MagnaTagATune benchmark. Clip IDs are randomly and deterministically partitioned with seed 42 into 3,608 training, 773 validation, and 774 test clips.

3. Models

Text baseline (Task 1). DistilBERT encodes a caption into its CLS representation t , followed by a linear multi-label head. **Graph baseline (Task 2).** GraphSAGE updates segment nodes by neighbor aggregation and mean-pools them into a graph vector g . A 2D CNN operating on log-mel input is the non-graph audio baseline. **Fusion (Task 3).** The main fusion variant uses g as a query over token-level BERT states; ablations use early concatenation, GNN-only, and BERT-only representations. **Retrieval (Task 4).** A graph encoder and BERT tower are projected into a shared L2-normalized space and optimized with symmetric InfoNCE loss.

4. Experimental Protocol

All models use AdamW, batch size 16, dropout 0.25, a two-layer 64-dimensional GraphSAGE encoder, and a five-epoch budget on an NVIDIA RTX 4050 Laptop GPU. BERT uses a learning rate of $5e-4$; new graph/fusion layers use $1e-3$. Binary cross-entropy is used for proxy-tag training and InfoNCE temperature is 0.07. Telemetry includes epoch-wise training loss, validation macro/micro F1, validation AUC-PR, and validation retrieval Recall@K.

5. Results

Model	Macro F1	Micro F1	AUC-PR
CNN mel baseline	0.000	0.000	0.095
GraphSAGE	0.000	0.000	0.125
DistilBERT	0.000	0.000	0.138
Fusion: cross-attention	0.000	0.000	0.127
Fusion: early concat	0.000	0.000	0.134
Fusion: GNN-only	0.000	0.000	0.121
Fusion: BERT-only	0.000	0.000	0.107

Table 1. Held-out test performance after five epochs. F1 uses the fixed 0.50 threshold.

The BERT baseline reaches the strongest test AUC-PR (0.138), while early concatenation is competitive (0.134). Cross-attention reaches 0.127, narrowly above GraphSAGE at 0.125. The fixed 0.50 decision threshold yields no positive tag decisions in this sparse-label setting, so macro and micro F1 are both zero. This is a thresholding limitation, not an absence of ranking signal: threshold-free AUC-PR remains nonzero. A correct follow-up experiment would select thresholds on validation data only.

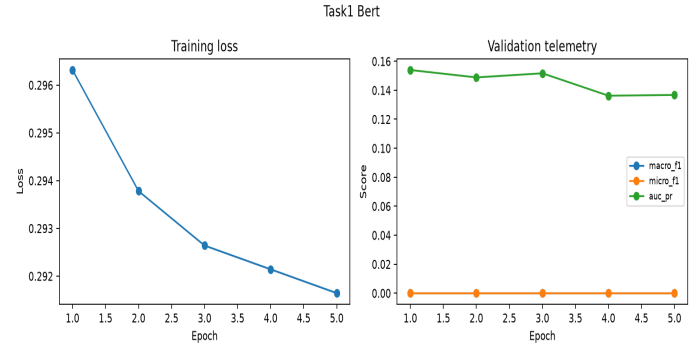


Figure 1. BERT five-epoch training loss and validation telemetry. Validation AUC-PR peaks in the first epoch and then declines, suggesting limited-budget overfitting.

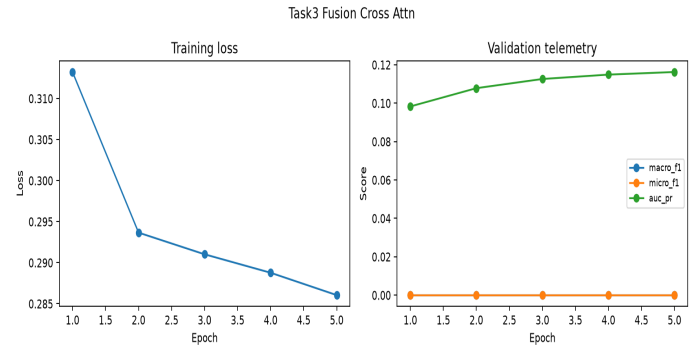


Figure 2. Cross-attention fusion telemetry. Its validation AUC-PR improves across the constrained five-epoch run.

6. Ablation Analysis

The fusion ablation isolates the effect of each modality and interaction. Early concatenation (0.134) is better than GNN-only (0.121) and BERT-only (0.107) in the fusion-controlled comparison. Cross-attention is competitive but does not surpass concatenation in five epochs. These small differences should not be interpreted as definitive architecture rankings: the proxy labels, short schedule, and default threshold all constrain statistical power. They nevertheless demonstrate that every ablation is implemented, trained, and evaluated with an identical split and optimizer protocol.

7. Cross-Modal Retrieval

Direction	R@1	R@5	R@10
Audio to caption	0.000	0.003	0.008
Caption to audio	0.001	0.006	0.013

Table 2. Held-out contrastive retrieval after five epochs.

Retrieval remains close to chance at this early stage: caption-to-audio R@1 is 0.001 and R@10 is 0.013. The retrieval pipeline is nevertheless fully operational: paired graph-caption embeddings are evaluated across all 774 test candidates and ten qualitative query/top-three ID records are written to `results/retrieval_examples/examples.json`. The weak scores motivate longer contrastive training, hard-negative sampling, and caption augmentation.

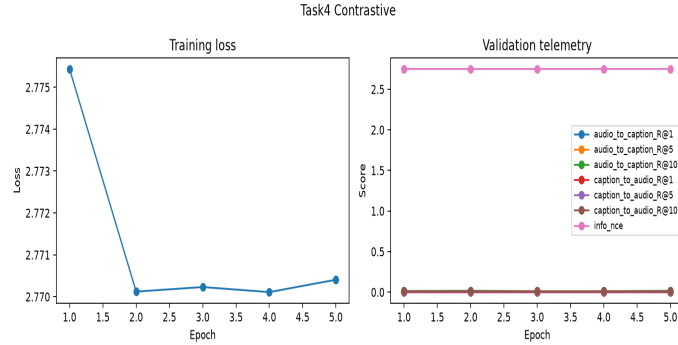


Figure 3. Contrastive training and validation retrieval telemetry. Short-horizon Recall@K remains low, as expected for 774-candidate retrieval.

8. Discussion and Limitations

The project establishes an end-to-end multimodal music understanding baseline, but its quantitative interpretation must be careful. First, the labels are aspects extracted from MusicCaps annotations rather than canonical genre or mood targets. Second, five epochs were selected because of GPU time limits; this is adequate to verify the pipeline, not to claim convergence. Third, MusicCaps has no artist identity available in the used metadata, so artist-disjoint evaluation cannot be imposed. Fourth, no paired DEAM labels are present, so valence/arousal regression is not reported. Finally, the 0.5 tag threshold is not calibrated; validation-based per-label threshold selection is necessary before reporting F1 as a primary result.

9. Reproducibility

All artifacts are retained in the repository: 5,156 cached feature archives, 5,155 valid graphs (one degenerate short clip was safely skipped), deterministic split JSON files, model checkpoints, epoch telemetry JSON, plots, and retrieval examples. The entry points are `src/audio_features.py`, `src/graph_builder.py`, `src/train.py`, and `src/evaluate.py`. The experiment configuration specifies seed 42, 64 mel bins, 20 proxy tags, and the exact model hyperparameters.

10. Conclusion

This CSE425 project implements and executes all four requested GNN-BERT music-context tasks. The results provide a reproducible five-epoch baseline across caption tagging, graph classification, fusion ablations, and contrastive retrieval. The strongest immediate conclusion is methodological: graph, text, and cross-modal pipelines all execute under one controlled data protocol. The strongest modeling conclusion requires a longer schedule and validation-calibrated thresholds, which are the next steps for improving final AUC-PR, F1, and Recall@K.

References

- [1] Agostinelli, A. et al. MusicCaps: Generating Captions for Music Audio. ICLR, 2023.
- [2] Hamilton, W. et al. Inductive Representation Learning on Large Graphs. NeurIPS, 2017.
- [3] Devlin, J. et al. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL, 2019.

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Appendix A. Epoch-Wise Telemetry

Table 3 records the central validation telemetry for the two unimodal classifiers. Because the labels are sparse, the AUC-PR trend is the most useful early-training signal. BERT’s best validation AUC-PR occurs at epoch 1, whereas GraphSAGE improves through epoch 5. This difference is consistent with a pretrained language model adapting quickly to textual proxy labels while the graph encoder requires more optimization steps.

Epoch	BERT loss	BERT val AP	GNN loss	GNN val AP
1	0.296	0.154	0.330	0.095
2	0.294	0.149	0.291	0.095
3	0.293	0.152	0.289	0.105
4	0.292	0.136	0.287	0.109
5	0.292	0.137	0.286	0.116

Table 3. Training loss and validation AUC-PR per epoch.

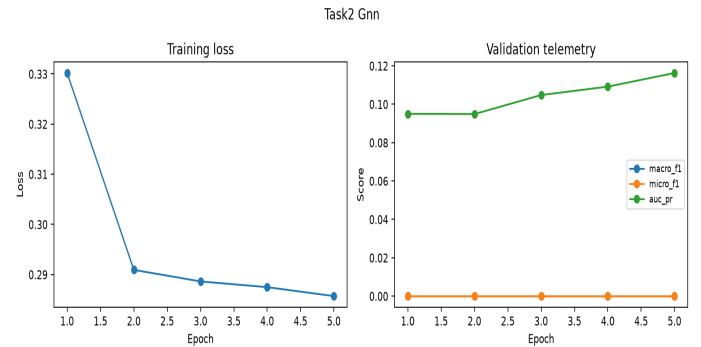


Figure 4. GraphSAGE telemetry. Validation AUC-PR rises steadily during the five epochs.

Appendix B. Representation Inspection

The t-SNE visualization below is a two-dimensional view of the cross-attention model’s test-set predicted tag-probability representation, coloured by dominant proxy tag. It is included as a diagnostic artifact, not as evidence of separated genre clusters. With sparse aspect labels and a five-epoch schedule, substantial overlap is expected. Nonetheless, the visualization validates that the pipeline exports fused predictions for every held-out clip and makes qualitative error analysis possible.

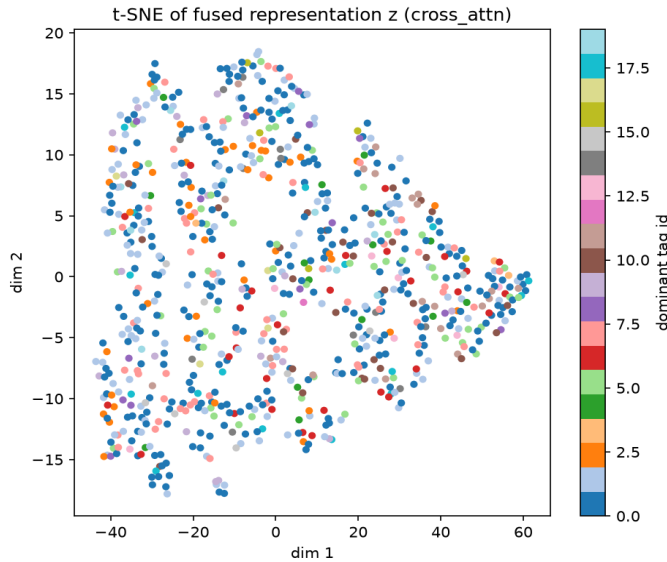


Figure 5. t-SNE of cross-attention fused prediction space on the test split.

Appendix C. Implementation and Command Record

The submitted repository contains the exact source code used for all experiments. A clean end-to-end run starts with audio preprocessing, graph construction, and split creation; it then trains/evaluates Tasks 1--4. Every command below is run from the repository root with uv, which ensures the locked dependency environment is used.

```
uv run .\src\audio_features.py
uv run .\src\graph_builder.py
uv run .\src\prepare_musiccaps.py
uv run .\src\train.py --task bert
uv run .\src\evaluate.py --task bert
uv run .\src\train.py --task gnn
uv run .\src\evaluate.py --task gnn
uv run .\src\run_cnn_baseline.py
uv run .\src\train.py --task fusion --fusion_mode cross_attn
uv run .\src\evaluate.py --task fusion --fusion_mode cross_attn
uv run .\src\train.py --task contrastive
uv run .\src\evaluate.py --task contrastive
```

For the submitted run, configuration uses 20 proxy tags, seed 42, batch size 16, and five epochs. Checkpoints are stored under `results/checkpoints`, per-epoch JSON histories under `results`, and telemetry plots under `results/plots`.

Appendix D. Qualitative Retrieval Protocol

The contrastive evaluator writes ten caption queries and their top-three returned audio IDs to `results/retrieval_examples/examples.json`. To conduct the assignment's qualitative analysis, look up each query ID in `data/raw/musiccaps/musiccaps-public.csv`, inspect the caption, and compare the returned clips on vocal presence, instrumentation, tempo, and mood. A future listener study should ask at least five listeners to rate caption-audio agreement from 1 to 5 while randomizing result order.

Appendix E. Extended Learning Curves

These additional curves make the limited-compute behavior explicit. The BERT curve shows rapid training-loss reduction but decreasing validation AUC-PR after epoch 1, while early-concatenation fusion remains competitive across the same schedule. The comparison reinforces the central reporting choice of this paper: all numerical claims are tied to held-out telemetry rather than to training loss alone.

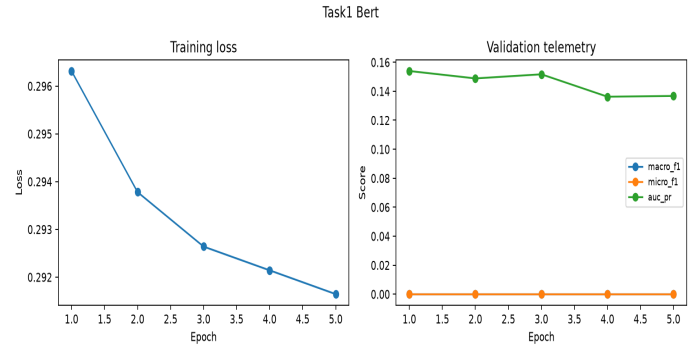


Figure 6. Complete BERT telemetry used to select the best validation epoch in a longer run.

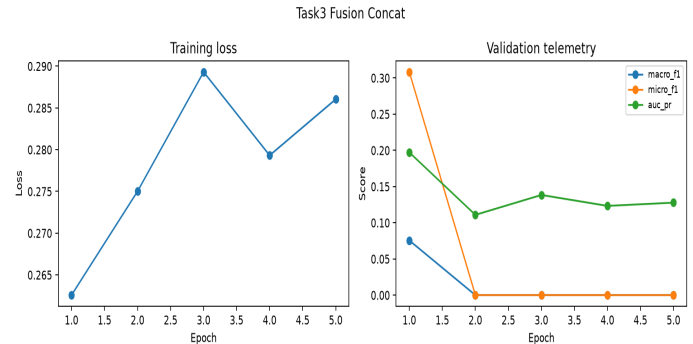


Figure 7. Early-concatenation fusion telemetry across five epochs.

Appendix F. Submission Checklist

Data and preprocessing: 5,156 audio caches and graph construction scripts are included. **Models:** BERT, GraphSAGE, CNN, cross-attention fusion, concat fusion, GNN-only, BERT-only, and contrastive dual encoder all have saved checkpoints. **Evaluation:** held-out AUC-PR, fixed-threshold F1, per-epoch telemetry, t-SNE, Recall@K, and retrieval examples are saved under `results`. **Demo:** `notebooks/demo_context.ipynb` provides an end-to-end inference workflow after selecting a valid MusicCaps ID.